■ Smart E-Commerce Analytics Platform

1-Month Capstone Project – Student Guidelines

Project Objective

The objective of this project is to build an end-to-end data science and machine learning analytics platform for an e-commerce company. Students will analyze business data, build multiple ML and DL models, and present insights using an interactive Streamlit dashboard.

Business Problems Addressed

1. Understand customer purchasing behavior
2. Segment customers into meaningful groups
3. Predict customer churn
4. Forecast future sales using LSTM deep learning
5. Recommend relevant products to customers
6. Present insights to business stakeholders through a dashboard

Datasets Used

1. Customers dataset – customer details and demographics
2. Transactions dataset – purchase history and sales data
3. Products dataset – product and category information

Project Timeline

1. Week 1: Data loading, cleaning, and exploratory data analysis (EDA)
2. Week 2: Customer segmentation (RFM + K-Means) and churn prediction
3. Week 3: Sales forecasting using LSTM and product recommendation system
4. Week 4: Streamlit dashboard development, report writing, and presentation

Models to be Built

1. K-Means clustering for customer segmentation
2. Logistic Regression or Random Forest for churn prediction
3. LSTM deep learning model for sales forecasting
4. Popularity-based or collaborative filtering recommendation system

Tools & Technologies

1. Python, Pandas, NumPy, Matplotlib, Seaborn
2. scikit-learn for machine learning
3. TensorFlow / Keras for LSTM
4. Streamlit for dashboard development

Final Deliverables

1. EDA and modeling notebooks
2. Trained and saved ML/DL models

3. Interactive Streamlit dashboard
4. Project report (PDF)
5. 5–7 minute project presentation

Expected Outcome: By completing this project, students will gain hands-on experience in solving real-world business problems using data science, machine learning, deep learning, and dashboard visualization.